# Integrated Single-Cell Transcriptomics, Cell-Cell Communication Inference, and Virtual Screening Prioritise CXCR4 and Natural Compound Candidates for Experimental Testing in Idiopathic Pulmonary Fibrosis


## Abstract

Idiopathic pulmonary fibrosis (IPF) is a progressive fatal disease with a median survival of 3–5 years and limited treatment options.[1,7] The CXCL12-CXCR4 axis is implicated in fibrotic remodelling, but computational prioritisation of natural CXCR4 inhibitors for IPF has not been reported.

Methods: We re-analysed single-cell RNA-sequencing data (GSE132771; n=3 IPF, n=3 normal donors),[4] inferred cell-cell communication using CellChat v2,[16,17] and performed structure-based virtual screening of 105 natural compounds against CXCR4 (PDB 3ODU) with AutoDock Vina (exhaustiveness=32).[18,19]

IPF lungs showed a 65% increase in fibroblasts and exclusive emergence of CTHRC1$^+$ myofibroblasts.[4,5] CellChat identified hyperactivation of the CXCL, SPP1, and MIF pathways (697 inferred interactions in IPF vs 502 in normal; +38.8%). Ten natural compounds (9.9% of the library) were predicted to bind CXCR4 with affinity better than AMD3100 (−9.471 kcal/mol), including Coptisine (−10.560), Genistein (−9.973), Glycyrrhetinic acid (−9.876), and Licochalcone A (−9.645).

This in silico study computationally prioritises the CXCR4 axis and specific natural compounds for experimental validation. All findings are predictions and require confirmation.


## Introduction

Idiopathic pulmonary fibrosis (IPF) is a progressive interstitial lung disease with a median survival of 3–5 years.[1,7] Two FDA-approved drugs (pirfenidone, nintedanib) slow functional decline but do not reverse fibrosis. Updated clinical practice guidelines continue to highlight the unmet need for disease-modifying therapies.[2,3]

Aberrant cell-cell communication between fibroblasts, macrophages, and epithelial cells drives IPF pathogenesis.[4,5,6] Landmark scRNA-seq studies identified CTHRC1$^+$ pathological myofibroblasts as key drivers.[4,5] The CXCL12-CXCR4 axis facilitates fibrocyte recruitment and myofibroblast persistence.[8,9,10] AMD3100 (Plerixafor), an approved CXCR4 antagonist, shows anti-fibrotic effects in preclinical models, validating CXCR4 as a druggable target.[12,13,14]

Natural compounds offer a chemically diverse library for initial CXCR4 inhibition exploration. Here, we integrate three complementary computational layers – scRNA-seq re-analysis, CellChat v2 communication inference, and structure-based virtual screening of 105 natural compounds – to computationally prioritise CXCR4-directed natural compounds for subsequent experimental testing in IPF.

## Methods

### 3.1 Dataset and scRNA-seq Data Processing

Single-cell RNA sequencing data were obtained from the Gene Expression Omnibus (GEO) under accession number GSE132771. Three IPF donors (GSM3891627, GSM3891629, GSM3891631) and three normal (NML) donors (GSM3891621, GSM3891623, GSM3891625) were selected from the Adams et al. (2020) cohort.[4] Unfractionated CellRanger v3 outputs were used to preserve all lineages.

Seurat v5 objects were created with min.cells = 3 and min.features = 200.[15] Per-sample QC thresholds: nFeature_RNA 500–4,000, nCount_RNA < 20,000, percent mitochondrial reads < 10% (IPF1: nFeature < 6,000, nCount < 40,000 due to higher transcriptional complexity). Each donor was downsampled to 3,000 cells (set.seed(42)), yielding 17,847 cells after QC.

CCA-based integration with 2,000 variable features (VST selection) was performed. The first 20 PCs were selected by ElbowPlot inspection for UMAP embedding and clustering (resolution = 0.1), yielding 14 initial clusters.

### 3.2 Cell Type Annotation and Differential Expression

Cell types were annotated using canonical marker FeaturePlots: EPCAM, SFTPC, AGER, SCGB1A1 (epithelial); CLDN5, CCL21, PECAM1, EMCN (endothelial); COL1A2, LUM, PDGFRA, PDGFRB (mesenchymal); PTPRC, CD52, AIF1, TRBC2 (immune). Fourteen clusters were merged into eight final cell types. FindAllMarkers() (Wilcoxon rank-sum, min.pct = 0.25, logfc.threshold = 0.25) refined annotation. Fibroblast differential expression: FindMarkers() with |log2FC| > 0.5 and adj. p < 0.05 (Benjamini-Hochberg). Pathway enrichment: enrichR with GO Biological Process 2023 and KEGG 2021 Human databases.[28]

### 3.3 CellChat v2 Cell-Cell Communication Inference

Separate CellChat objects for CON and IPF were created using createCellChat() with CellChatDB.human.[16,17] The pipeline included subsetData(), identifyOverExpressedGenes(), computeCommunProb(type = ‘triMean’, raw.use = TRUE), filterCommunication(min.cells = 10), and aggregateNet(). A permutation test (10,000 random shuffles) assessed pathway rewiring significance. NMF patterns identified with identifyCommunicationPatterns() (CON: k = 4; IPF: k = 5).

### 3.4 Molecular Docking

The CXCR4 crystal structure (PDB: 3ODU, 2.5Å)[20] was prepared in PyMOL: water molecules and the co-crystallised ligand IT1t were removed; polar hydrogens and Gasteiger charges were assigned using OpenBabel v3.1 at pH 7.4. A library of 105 bioactive natural compounds with MMFF94-optimised 3D coordinates was assembled from PubChem. Four compounds were excluded prior to analysis (two cytotoxic agents: camptothecin, celastrol; one CNS-liable compound: harmaline; one duplicate), leaving 101 compounds for scoring. Docking was performed with AutoDock Vina 1.2.5[18,19] using a 30 × 30 × 30 Å grid box centred on the IT1t binding pose (X = 20.610, Y = −7.972, Z = 71.068 Å), exhaustiveness = 32, nine binding modes per compound. AMD3100 (Plerixafor; CID 65015) served as the positive control[12]; pirfenidone (CID 40632) as the negative control. Lipinski Rule-of-Five compliance was assessed for all hits.[25]

### 3.5 Network Pharmacology

A PPI network for eight hub proteins (CD44, CXCR4, MIF, SPP1, CXCL12, CD74, ITGB1, CCL2) was constructed from STRING v12.0 (confidence ≥ 0.700).[21] Hub scores = sum of z-scores of degree, betweenness, and closeness centrality. Functional enrichment was performed via STRING built-in modules.

## Results

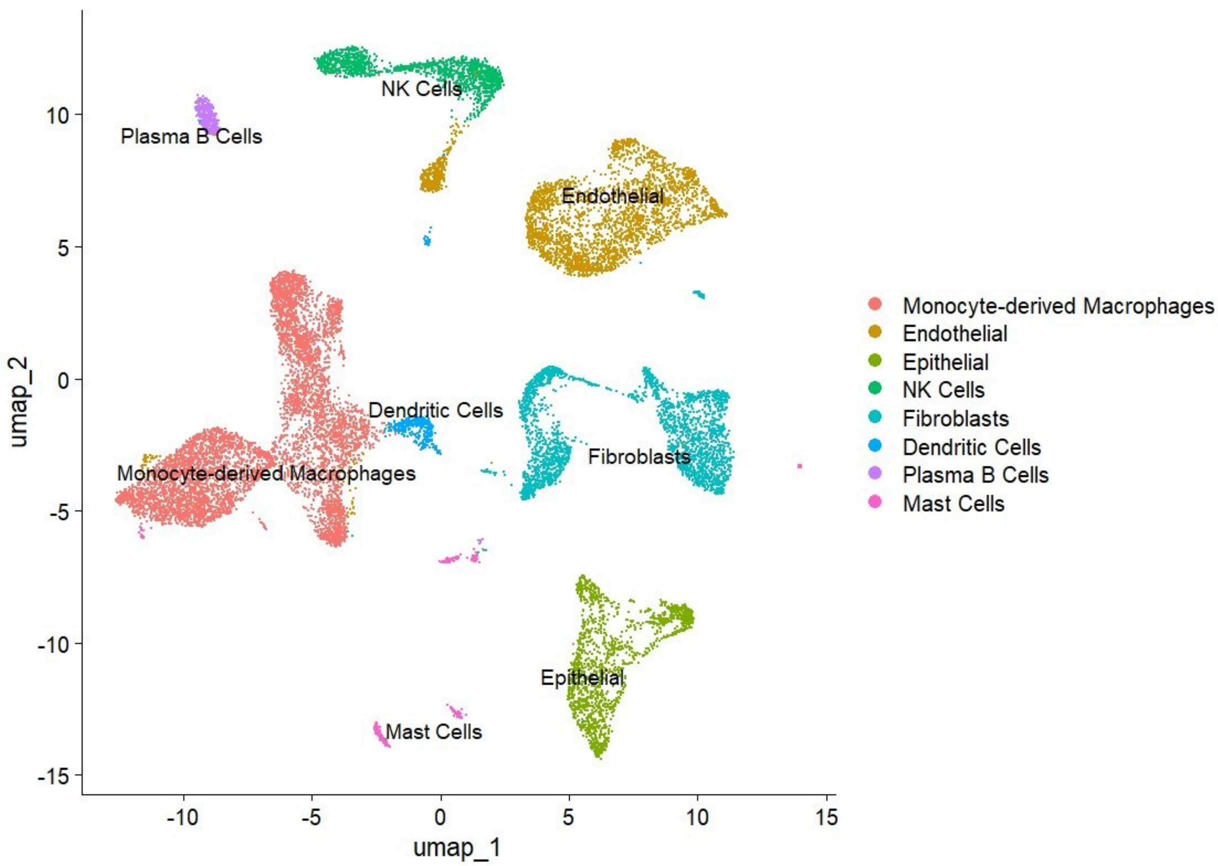
### 4.1 Single-Cell Transcriptomic Landscape of IPF and Normal Human Lung

#### 4.1.1 Quality Control, Integration, and Cell Type Annotation

After per-sample QC, 17,847 cells passed filtering (mean 2,974 per donor). CCA-based integration produced well-structured UMAP co-embedding of CON and IPF cells (Figure 1B), confirming successful batch correction.[15] Clustering at resolution 0.1 identified 14 clusters annotated into eight cell types (Figure 1A, 1C–1D): Monocyte-derived Macrophages (S100A8, FCN1), Endothelial (CLDN5, PECAM1), Epithelial (EPCAM, SFTPC; AT2-dominant), NK Cells (KLRD1, PRF1), Fibroblasts (COL1A2, LUM), Dendritic Cells (CD1C, FCER1A), Plasma B Cells (IGHG1, IGKC), and Mast Cells (TPSB2). The canonical marker DotPlot confirmed high specificity for all eight cell types (Figure 1D).

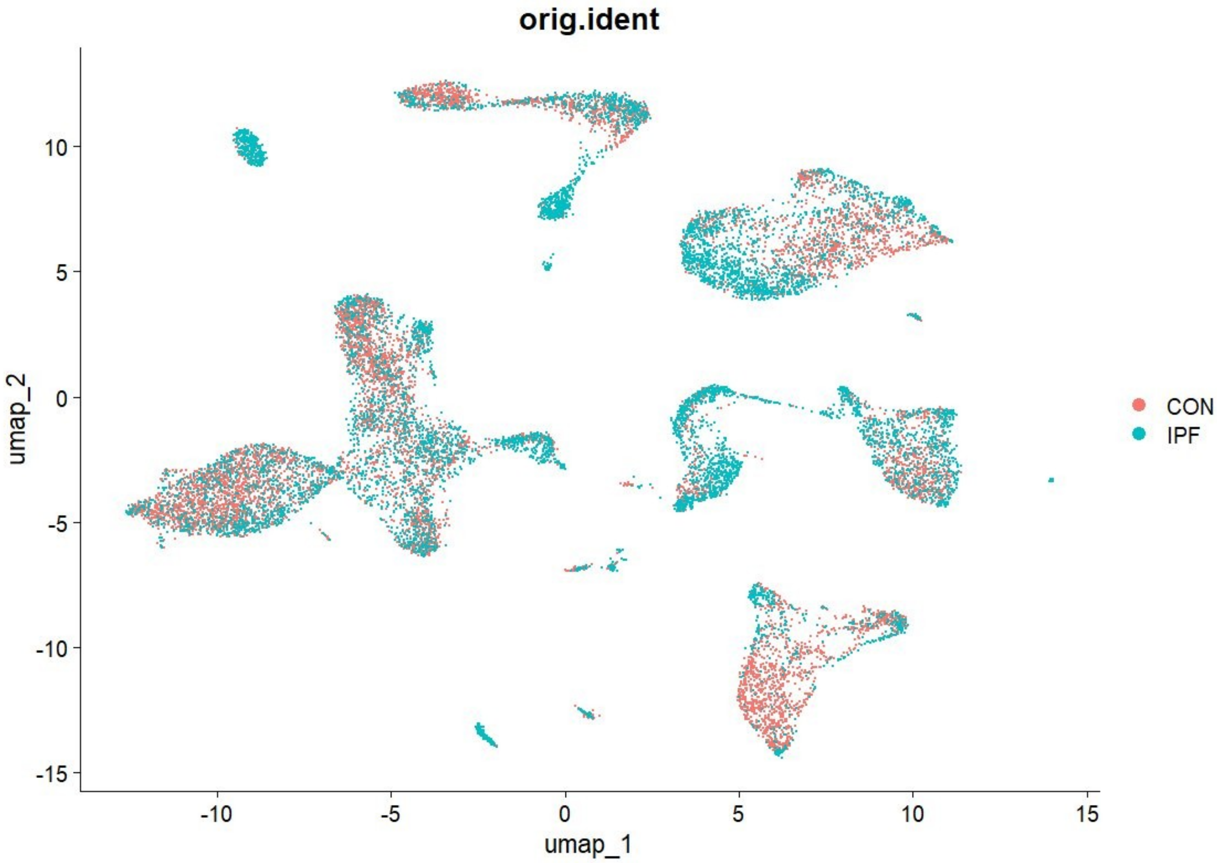
**Figure 1. Single-Cell Transcriptomic Landscape of IPF and Normal Human Lung**

**A – Annotated UMAP (CON + IPF combined)**



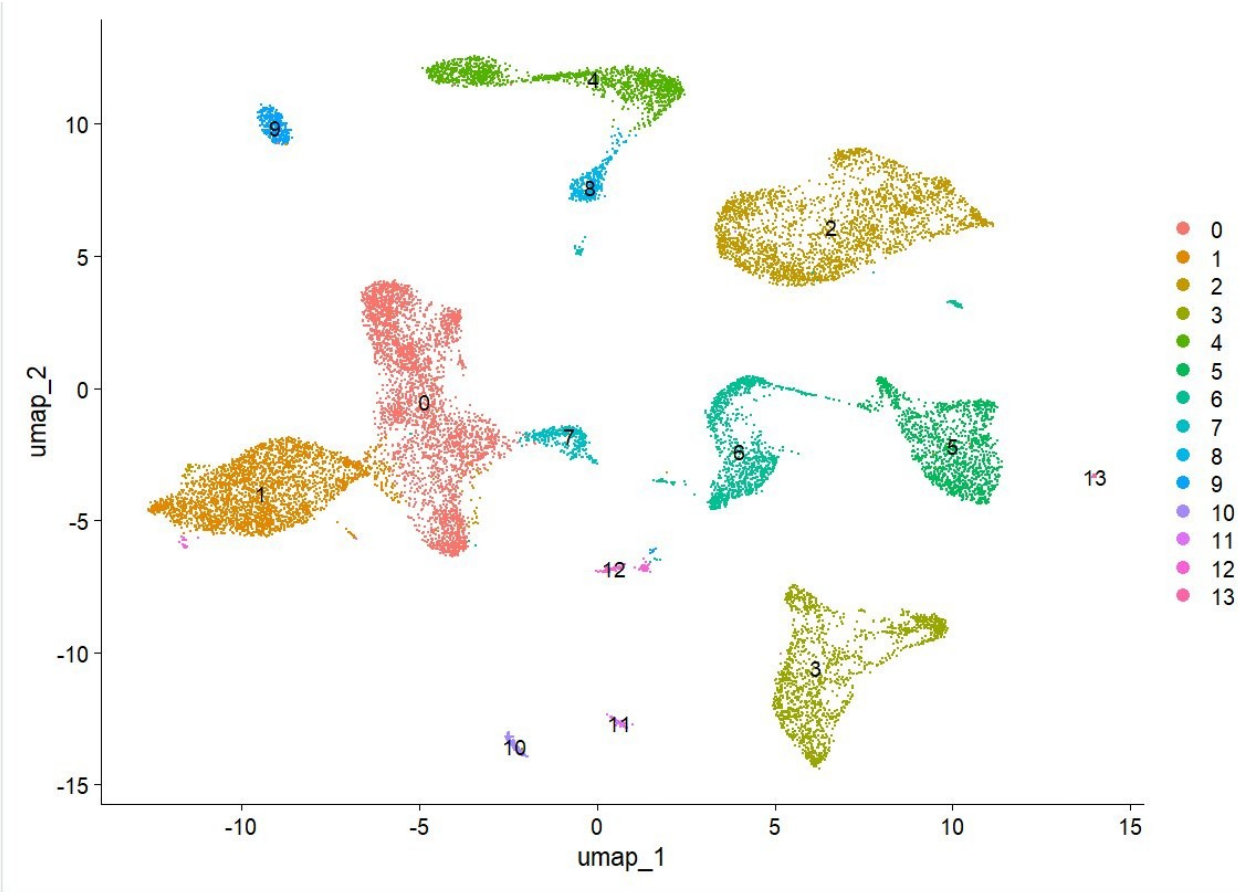

*(A) UMAP of 17,847 integrated cells annotated into eight final cell types after CCA-based integration.*

**B – UMAP coloured by condition**



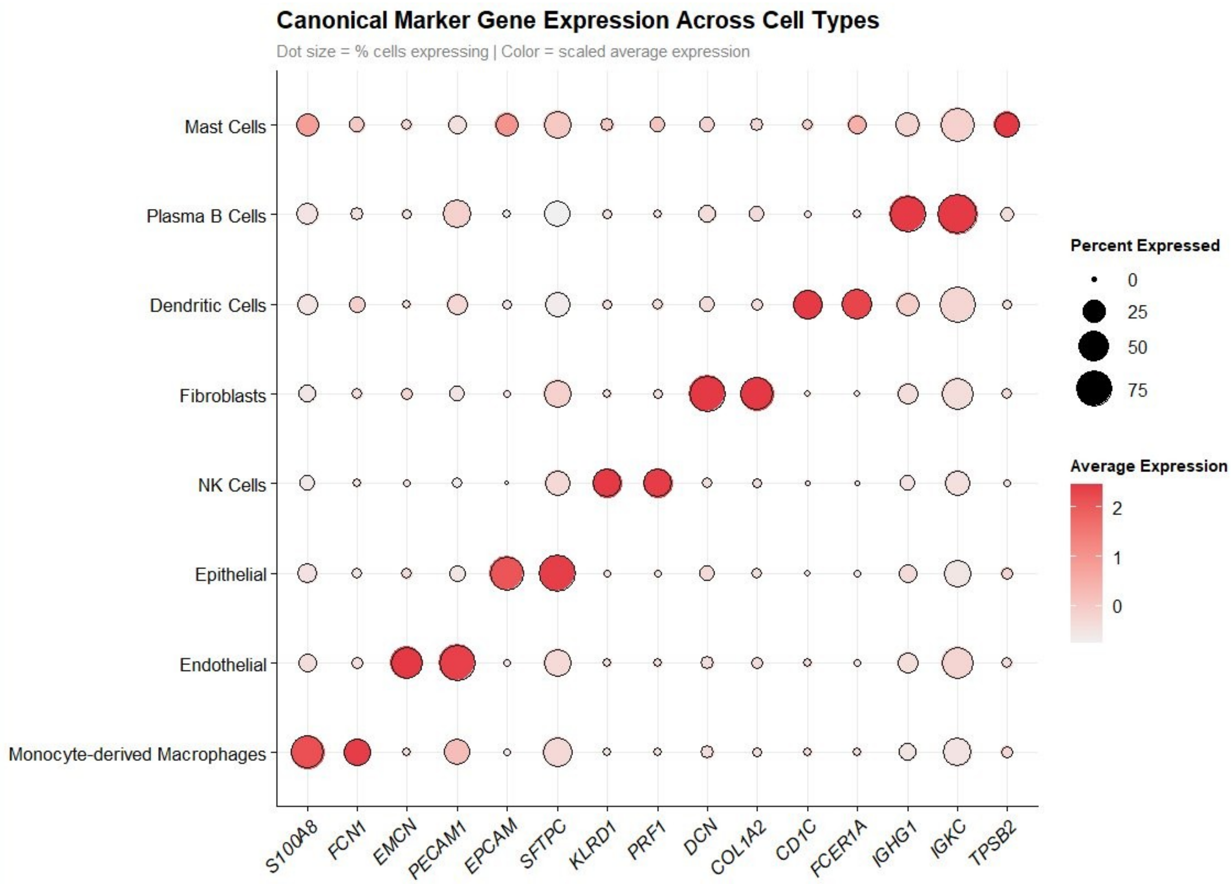

*(B) UMAP coloured by condition (CON = salmon, IPF = teal). Substantial co-embedding across clusters confirms successful batch correction.*

**C – Initial Seurat clusters (resolution = 0.1)**



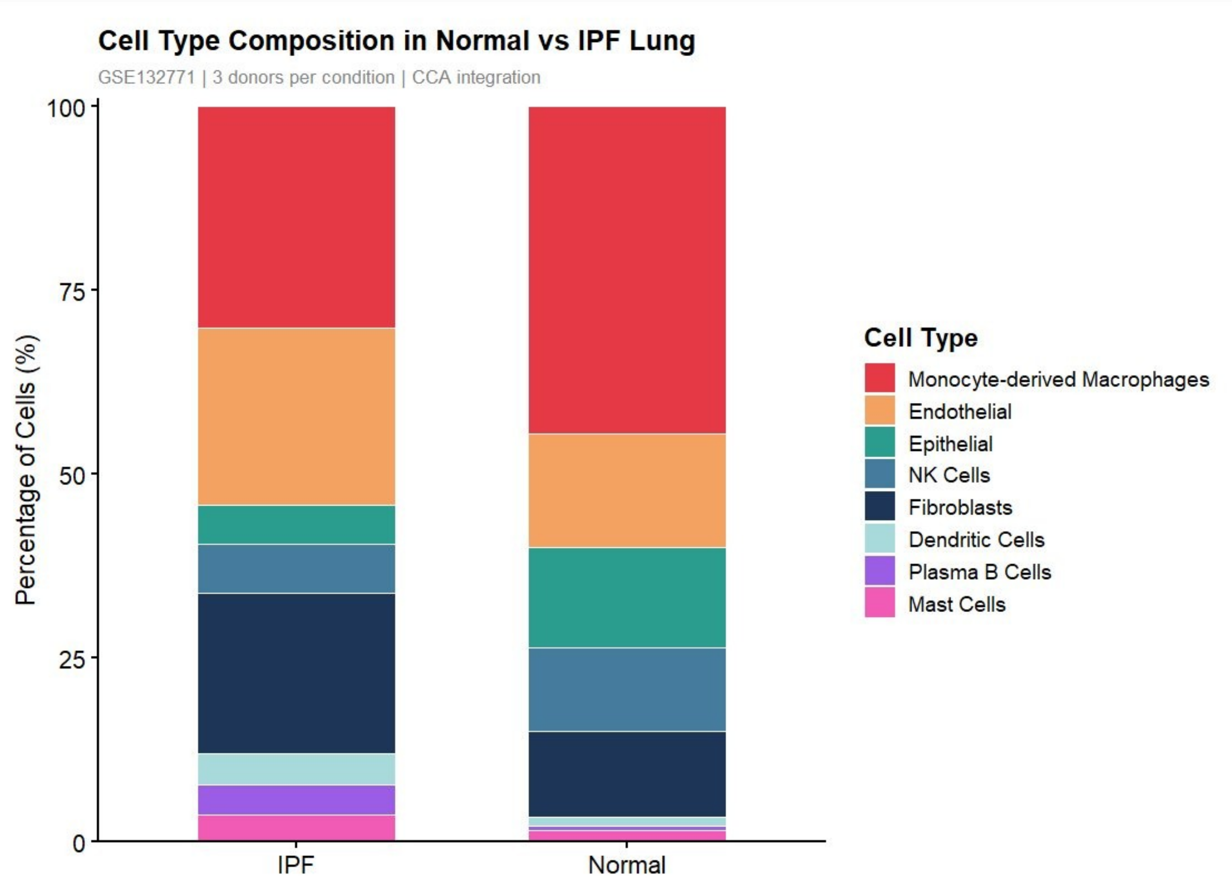

*(C) Fourteen initial Seurat clusters prior to manual annotation.*

**D – Canonical Marker DotPlot**




*(D) DotPlot of canonical markers across the eight annotated cell types. Dot size = % expressing cells; colour intensity = scaled average expression.*

### E – Cell Type Composition: Normal vs IPF




*(E) Stacked bar chart of cell type proportions. Fibroblasts increased from 12.2% to 20.1% (+65%); epithelial cells declined from 16.1% to 5.3% (−67%). Proportions are descriptive (n = 3 per condition).*

### 4.1.2 Disease-Associated Compositional Remodelling

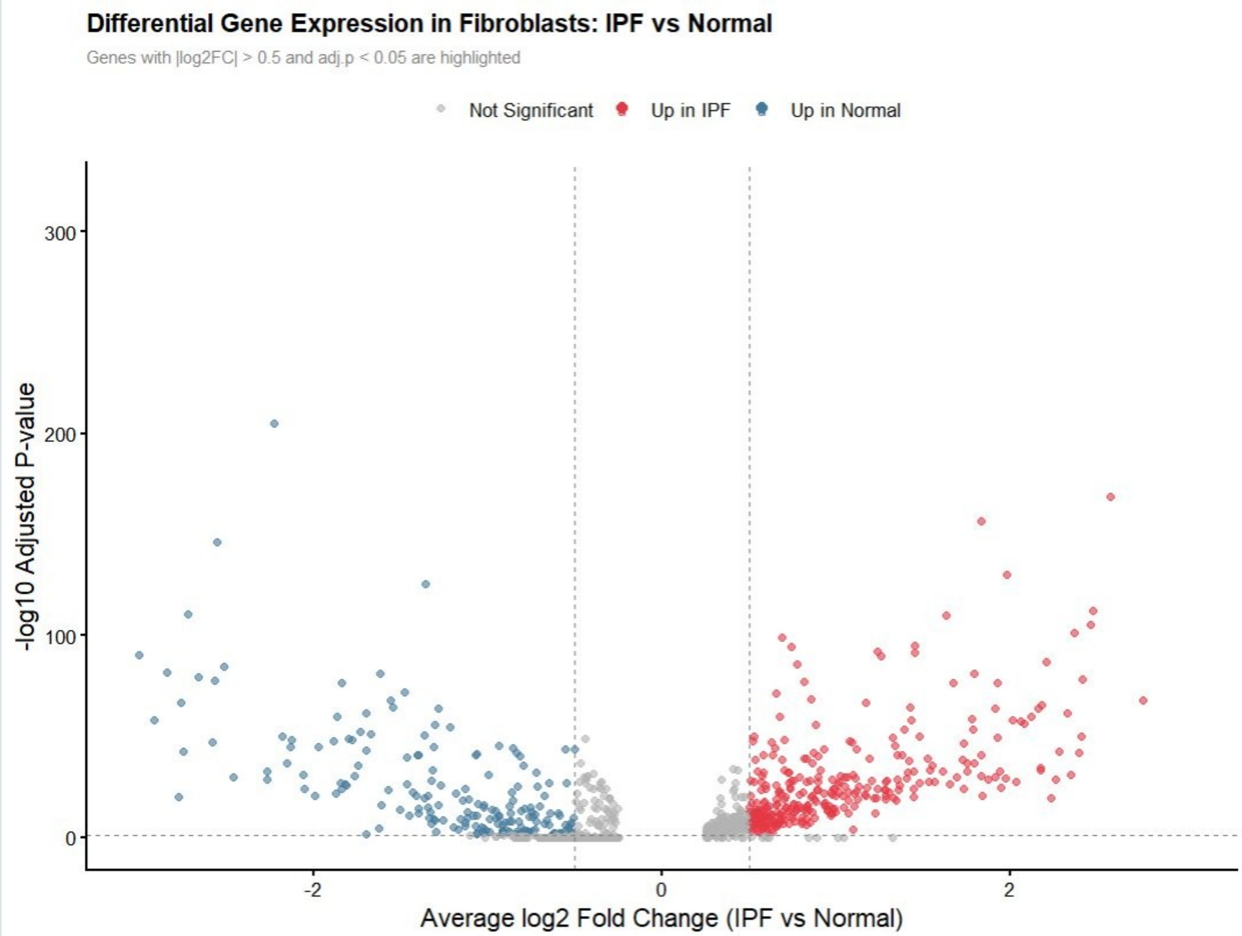
Comparison of cell type proportions (Figure 1E) revealed notable compositional remodelling. Fibroblast proportion increased from 12.2% to 20.1% (+65% relative), while epithelial proportion declined from 16.1% to 5.3% (−67% relative), consistent with alveolar destruction and fibroblastic replacement.[4,5] Dendritic cells and plasma B cells each increased from ~1% to ~5% (+400%), indicating enhanced adaptive immune activation. A plasma B cell cluster showed near-exclusive IPF enrichment, consistent with B cell accumulation reported in IPF bronchoalveolar lavage.

### 4.1.3 CTHRC1⁺ Pathological Myofibroblast and Fibroblast Differential Expression

CTHRC1 FeaturePlot analysis revealed exclusive expression in the IPF fibroblast compartment and a subset of IPF endothelial cells, absent in normal lung (Figure 2C), directly replicating the Adams et al. (2020) landmark finding[4] and suggesting endothelial-to-mesenchymal transition in IPF endothelium. Differential expression analysis within fibroblasts identified 189 significantly upregulated and 134 downregulated genes ($|\log2FC| > 0.5$, adj. $p < 0.05$; Figure 2A). GO Biological Process enrichment of IPF-upregulated fibroblast genes identified Regulation of Apoptotic Process, Supramolecular Fiber Organisation, ECM Organisation, Collagen Fibril Organisation, and Response to Unfolded Protein (Figure 2D) – all mechanistically coherent with established IPF biology.[4,5,6]

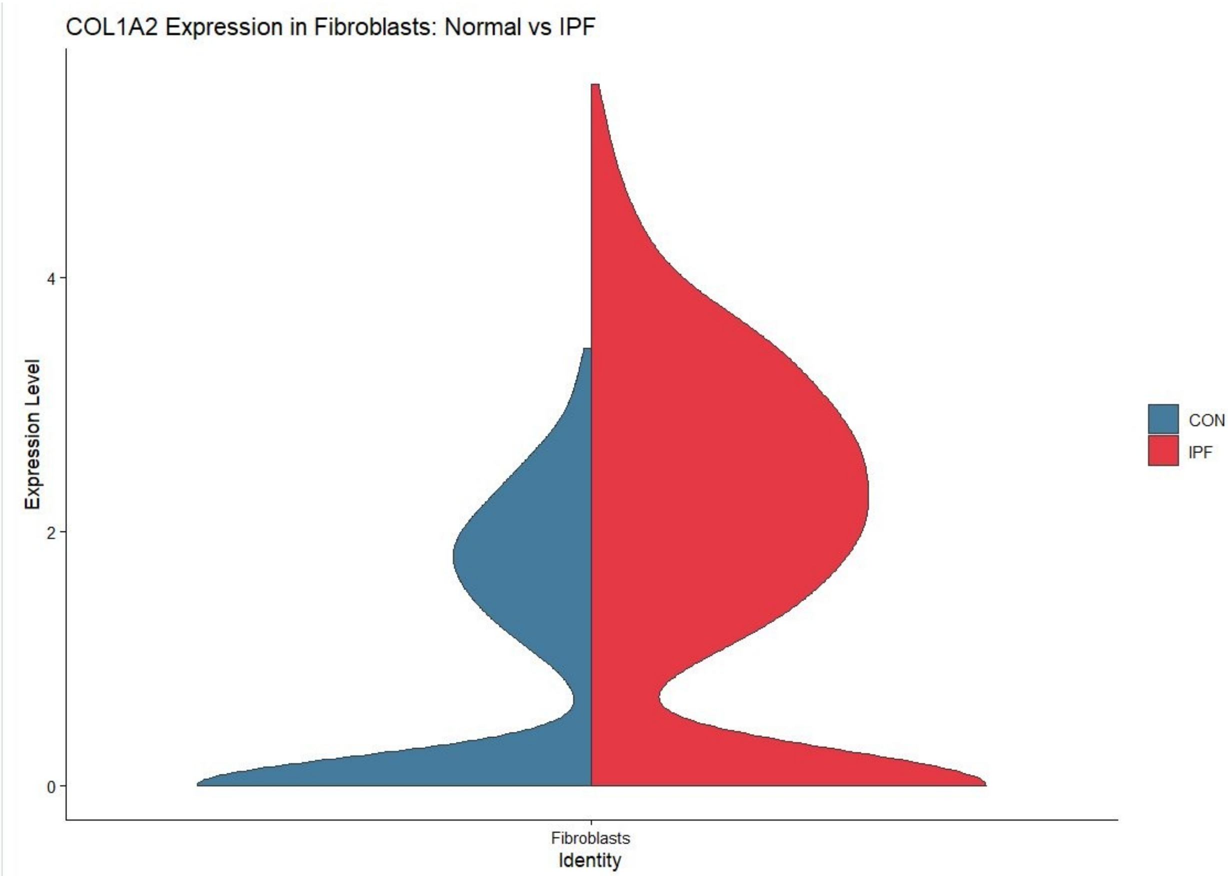
## Figure 2. Fibroblast-Specific Transcriptomic Changes in IPF

### A – Volcano Plot: Fibroblast Differential Expression (IPF vs Normal)



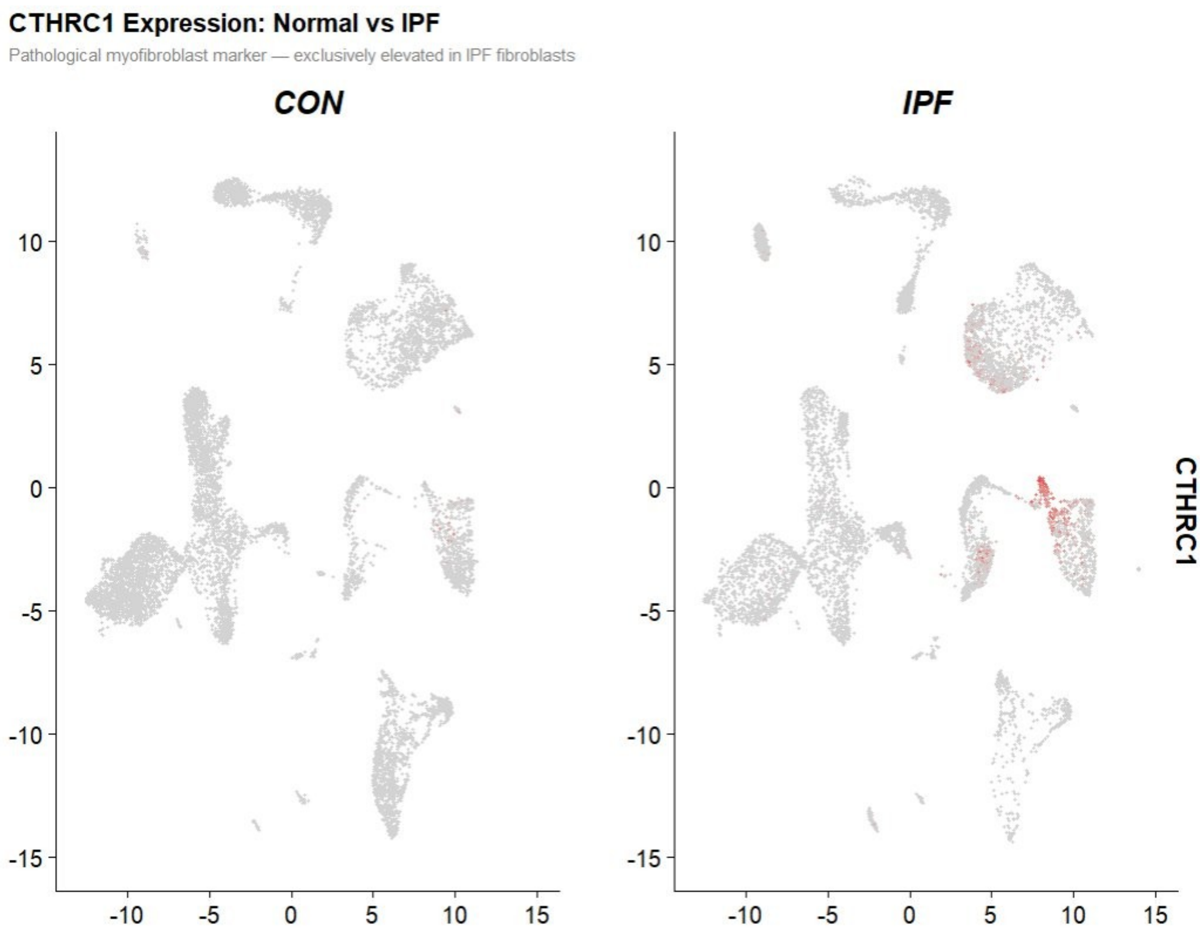

*(A) Volcano plot of fibroblast differential expression. Red = upregulated in IPF (n = 189); blue = upregulated in normal (n = 134). Thresholds: |log2FC| > 0.5, adj. p < 0.05.*

### B – COL1A2 Expression in Fibroblasts: Normal vs IPF



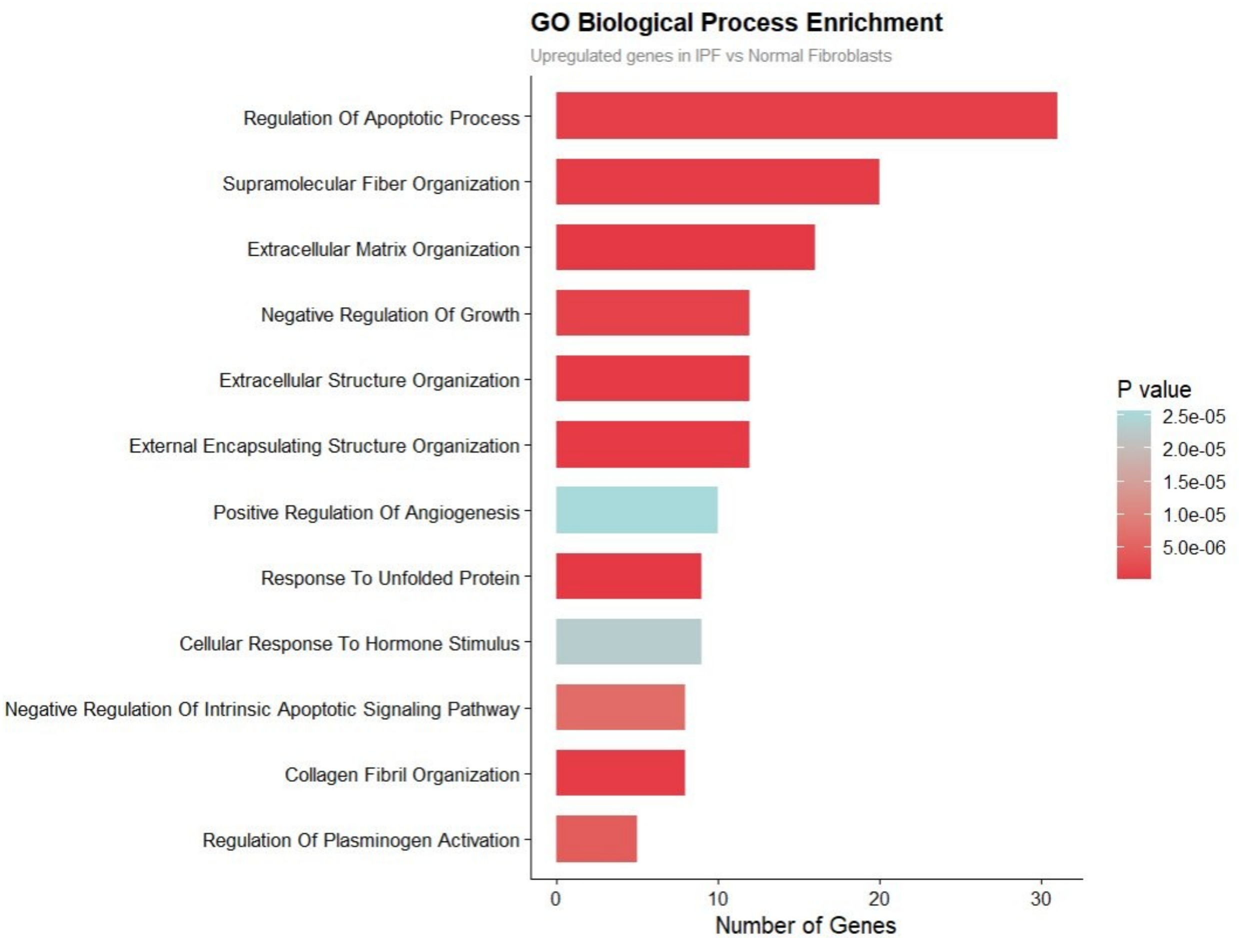

*(B) COL1A2 violin plot in fibroblasts. Bimodal IPF distribution (peak ~5.5 vs ~3.2 in normal) indicates co-existence of quiescent and highly activated COL1A2-high myofibroblast subpopulations.*

**C – CTHRC1 Expression: Normal vs IPF**




*(C) CTHRC1 FeaturePlot split by condition. Exclusively elevated in IPF fibroblasts and endothelial cells; absent in normal lung. Replicates Adams et al. (2020)[4] and is consistent with endothelial-to-mesenchymal transition.*

**D – GO Biological Process Enrichment: IPF Fibroblast Upregulated Genes**




*(D) Top GO Biological Process terms. ECM organisation, apoptosis regulation, and unfolded protein response reflect the three pillars of IPF pathogenesis.*

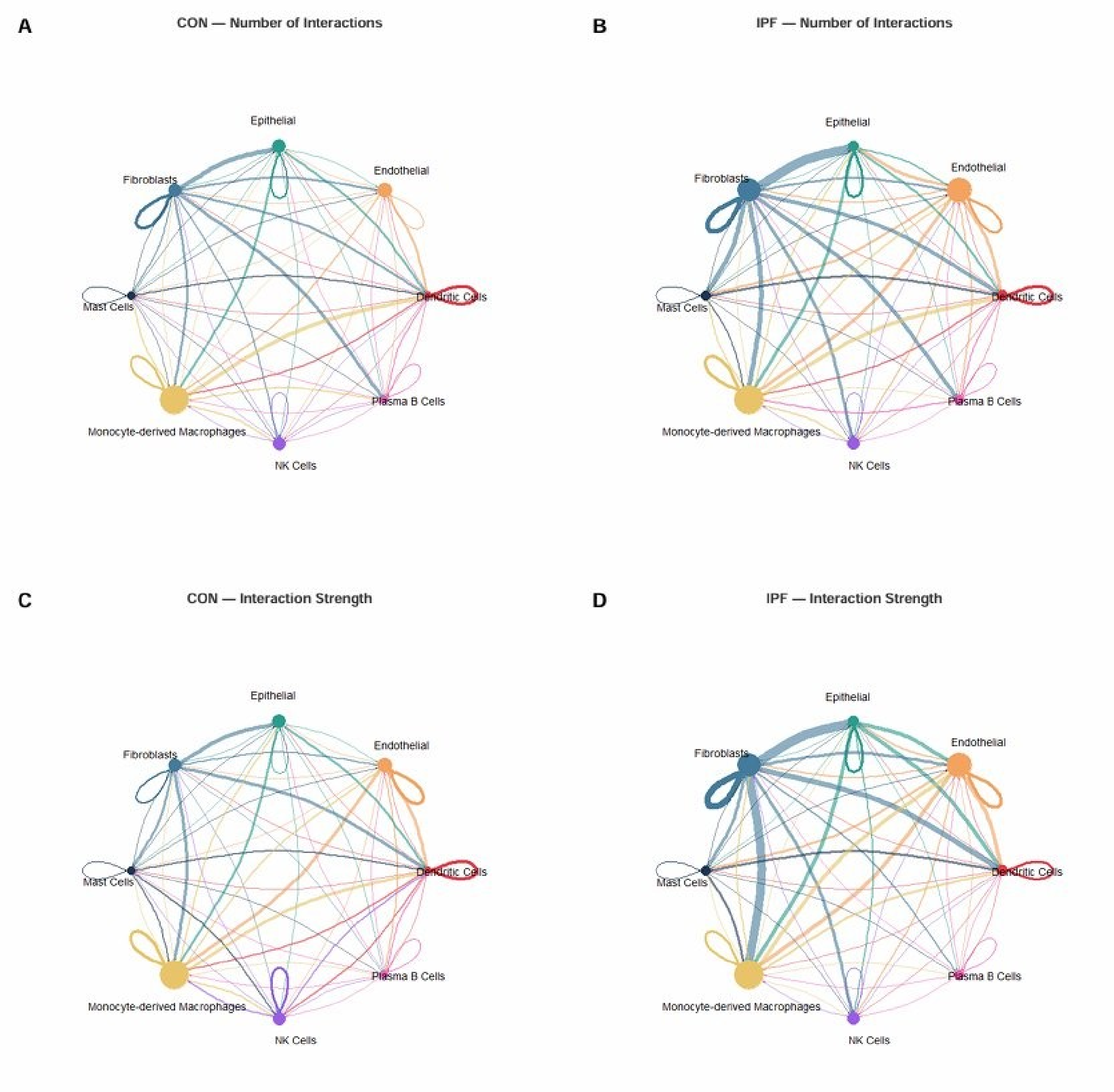
### 4.2 CellChat v2: Hyperactivated, Fibroblast-Centred Communication Network in IPF

CellChat v2[16,17] inferred 697 significant ligand-receptor interactions in IPF vs 502 in normal (+38.8%), with total interaction strength increasing from 8.812 to 12.046 (+36.7%; Figure 3A). Information flow analysis identified 10 gained and 12 lost pathways in IPF (Figure 3B). Top gained pathways: SPP1 (10.0),[26,27] CCL (9.1), CXCL (9.0),[8,9,10] HGF (8.0), FGF (7.0), THY1 (7.0). Top lost pathways: IFN-II (8.1), IL10 (7.5), OCLN (6.0) – representing dismantled immune surveillance and epithelial barrier integrity.

The fibroblast self-loop (collagen-integrin and fibronectin-CD44 signalling) emerged as the dominant new interaction in IPF (Figure 3A), indicating that autocrine fibroblast sustenance – absent in normal lung – becomes the dominant communication feature in disease. Fibroblasts were the dominant senders (outgoing strength 4.32) and monocyte-derived macrophages the dominant receivers (incoming strength 2.78). The bubble plot (Figure 3C; $p < 0.01$) confirmed CXCL2/3/8 → ACKR1 as the highest-probability new IPF interactions (~0.08), with MIF → CD74/CXCR4 co-receptor engagement prominent.

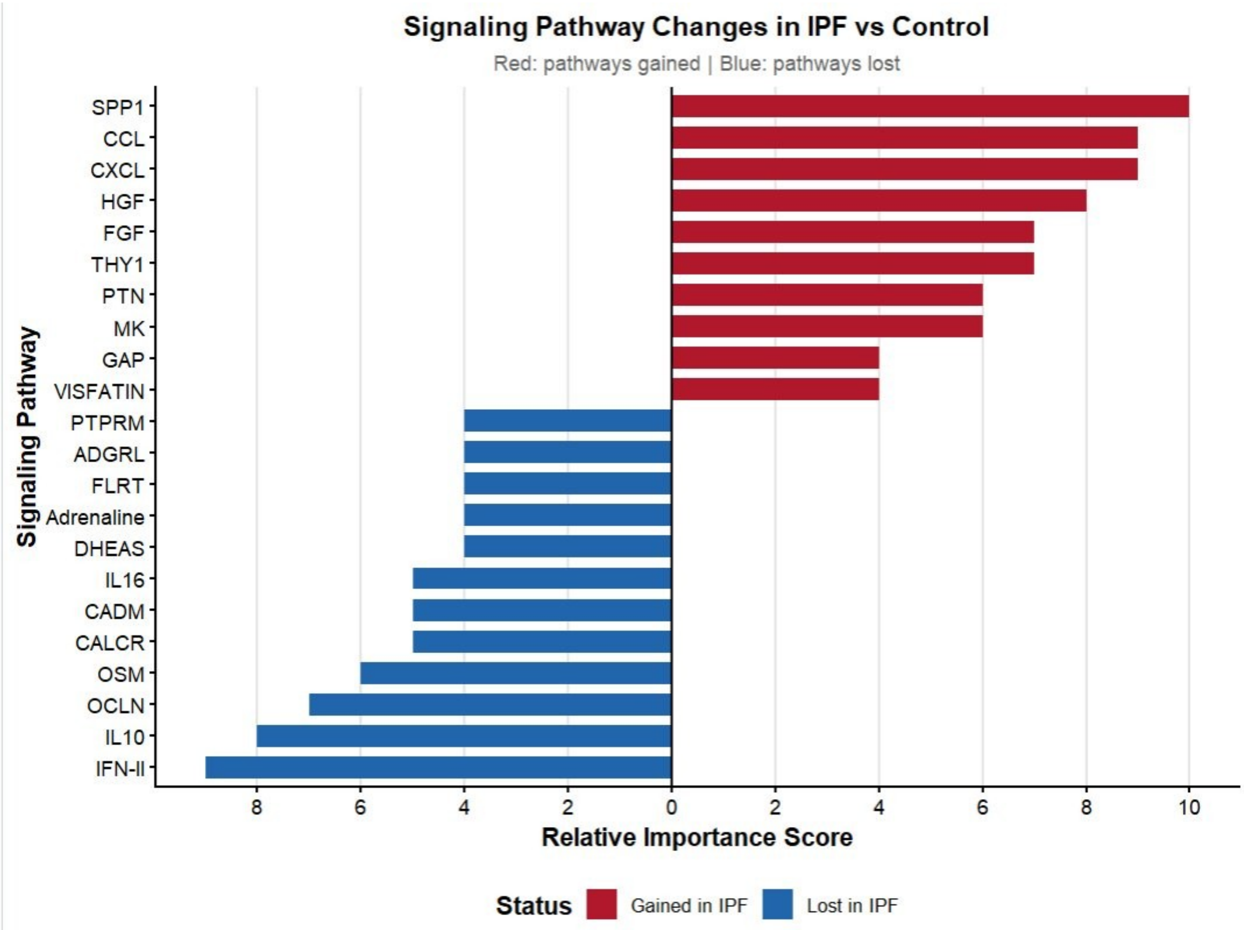
**Figure 3. CellChat v2 Cell-Cell Communication Analysis: IPF vs Normal Lung**

**A – Circle Plots: Number of Interactions and Interaction Strength**



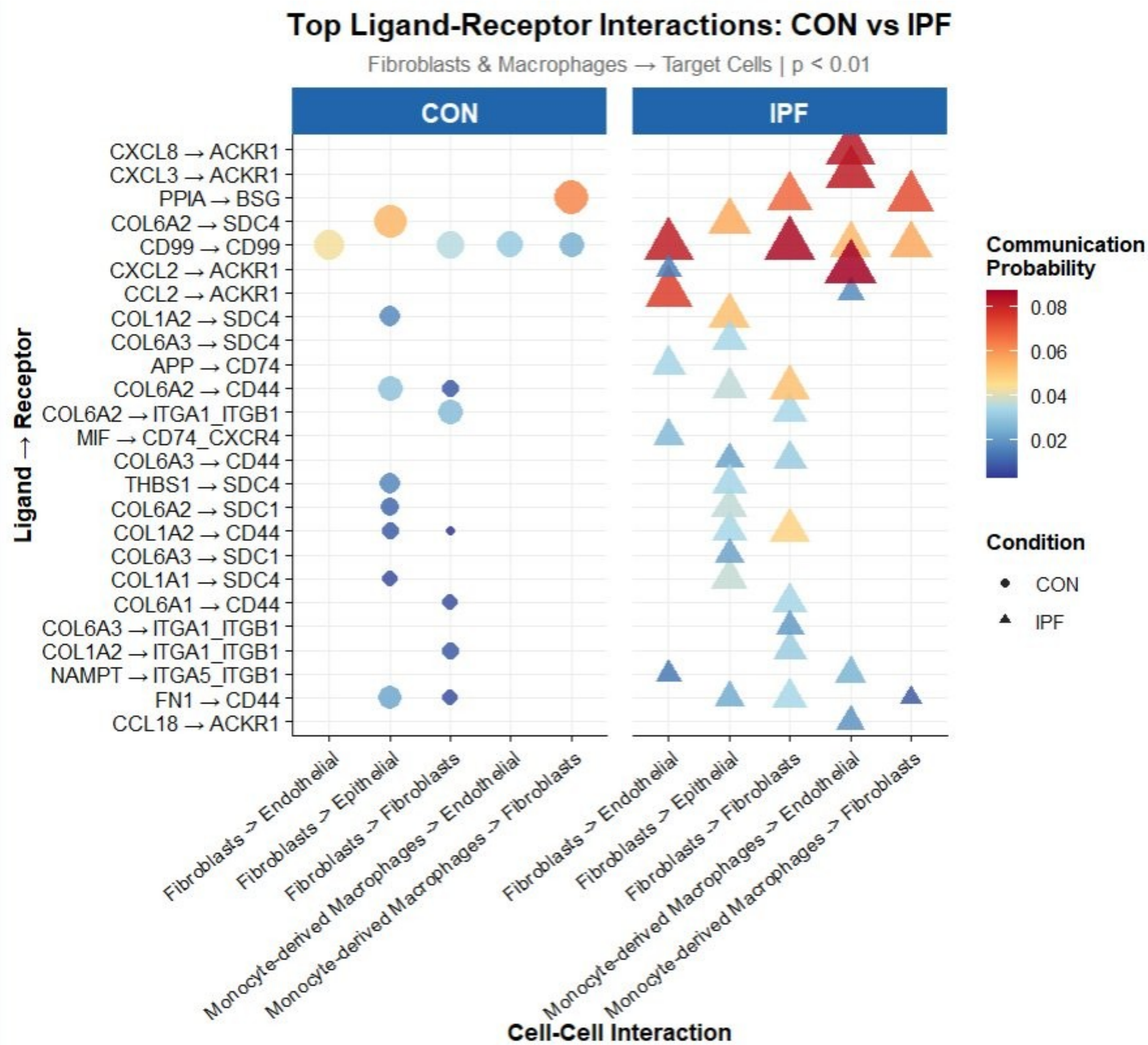

*(A) Circle plots for CON (A, C) and IPF (B, D) showing number of interactions and interaction strength. The dominant fibroblast self-loop and expanded fibroblast–macrophage axis are hallmarks of IPF pathology.*

**B – Signalling Pathway Changes in IPF vs Control**




*(B) Gained (red) and lost (blue) signalling pathways ranked by relative importance score. SPP1, CCL, and CXCL are most prominently gained; IFN-II, IL10, and OCLN are lost.*

**C – Top Ligand-Receptor Interactions: CON vs IPF**




*(C) Bubble plot of top ligand-receptor interactions from fibroblasts and macrophages (p < 0.01). CXCL2/3/8 → ACKR1 show the highest communication probability in IPF (triangles ~0.08); MIF → CD74_CXCR4 engagement is also prominent.*

### 4.3 Structure-Based Virtual Screening Identifies Natural Compound CXCR4 Binders

AMD3100 docked at −9.471 kcal/mol, consistent with published Vina scores against 3ODU,[18,19] validating the grid box. Pirfenidone scored −6.098 kcal/mol, confirming weak non-specific CXCR4 interaction as expected.

Score distribution across 101 compounds (Figure 5A) showed a left-skewed profile (mode: −8.0 to −8.5 kcal/mol). Ten compounds (9.9%) surpassed the AMD3100 threshold. After exclusion of cytotoxic agents and CNS-liable compounds, four were prioritised (Table 1). All four pass Lipinski Rule-of-Five criteria[25] and have established pharmacological safety profiles in experimental models.

**Table 1. Top-Prioritised CXCR4 Hits and Controls**

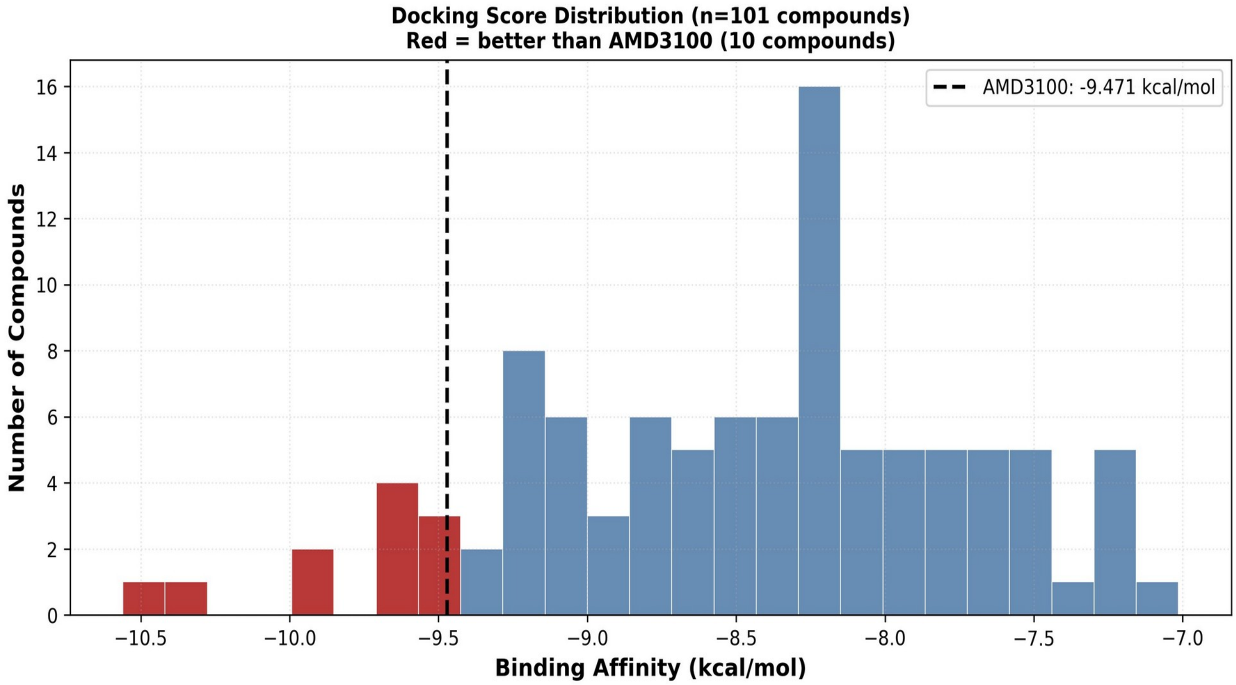
*Bold compound names exceed AMD3100 threshold (−9.471 kcal/mol). Green = hits; yellow = controls. Convergence: HIGH = spread < 2.0 kcal/mol with modes 1–3 within 0.5 kcal/mol of mode 1.*

| Compound | CID | Affinity (kcal/mol) | MW (Da) | cLogP | RO5 | Convergence |
|---|---|---|---|---|---|---|
| **Coptisine** | 72322 | -10.560 | 320.2 | 2.81 | Pass | MEDIUM |
| **Genistein** | 65752 | -9.973 | 270.2 | 3.10 | Pass | MEDIUM |
| **Glycyrrhetinic acid** | 265237 | -9.876 | 470.4 | 3.35 | Pass | MEDIUM |
| **Licochalcone A** | 5316743 | -9.645 | 338.4 | 4.12 | Pass | HIGH |
| AMD3100 [positive ctrl] | 65015 | -9.471 | 502.5 | 0.42 | Borderline | — |
| Pirfenidone [negative ctrl] | 40632 | -6.098 | 185.2 | 0.90 | Pass | — |

Binding pose analysis (Figure 5B–5E): Coptisine – planar isoquinoline core superimposes on the IT1t chromenone ring with π-stacking at Trp94/Tyr116 and polar contacts toward Asp171, Asp262, Glu288; no prior CXCR4 activity has been described for this compound. Genistein – A-ring hydroxyls engage Asp171/Glu288; consistent with published experimental CXCR4 inhibitory data in cancer cell migration assays.[22] Glycyrrhetinic acid – pentacyclic scaffold in the TM4-TM5 hydrophobic channel; 11-oxo group H-bonds with Tyr116; anti-fibrotic activity in IPF models reported independently.[23] Licochalcone A – highest pose convergence (HIGH; spread = 1.676 kcal/mol); α,β-unsaturated ketone near Cys109 in extracellular loop 1; attenuates TGF-β1-induced fibroblast-to-myofibroblast transition.[24]

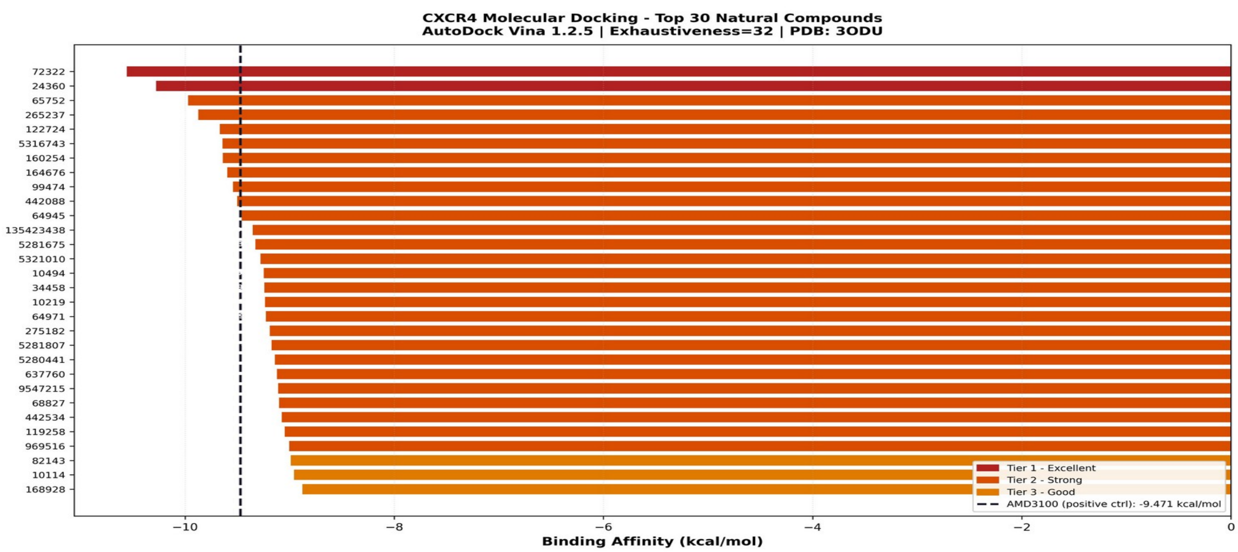
## Figure 4. Structure-Based Virtual Screening Against CXCR4 (PDB: 3ODU)

### A – Docking Score Distribution (n = 101 compounds)



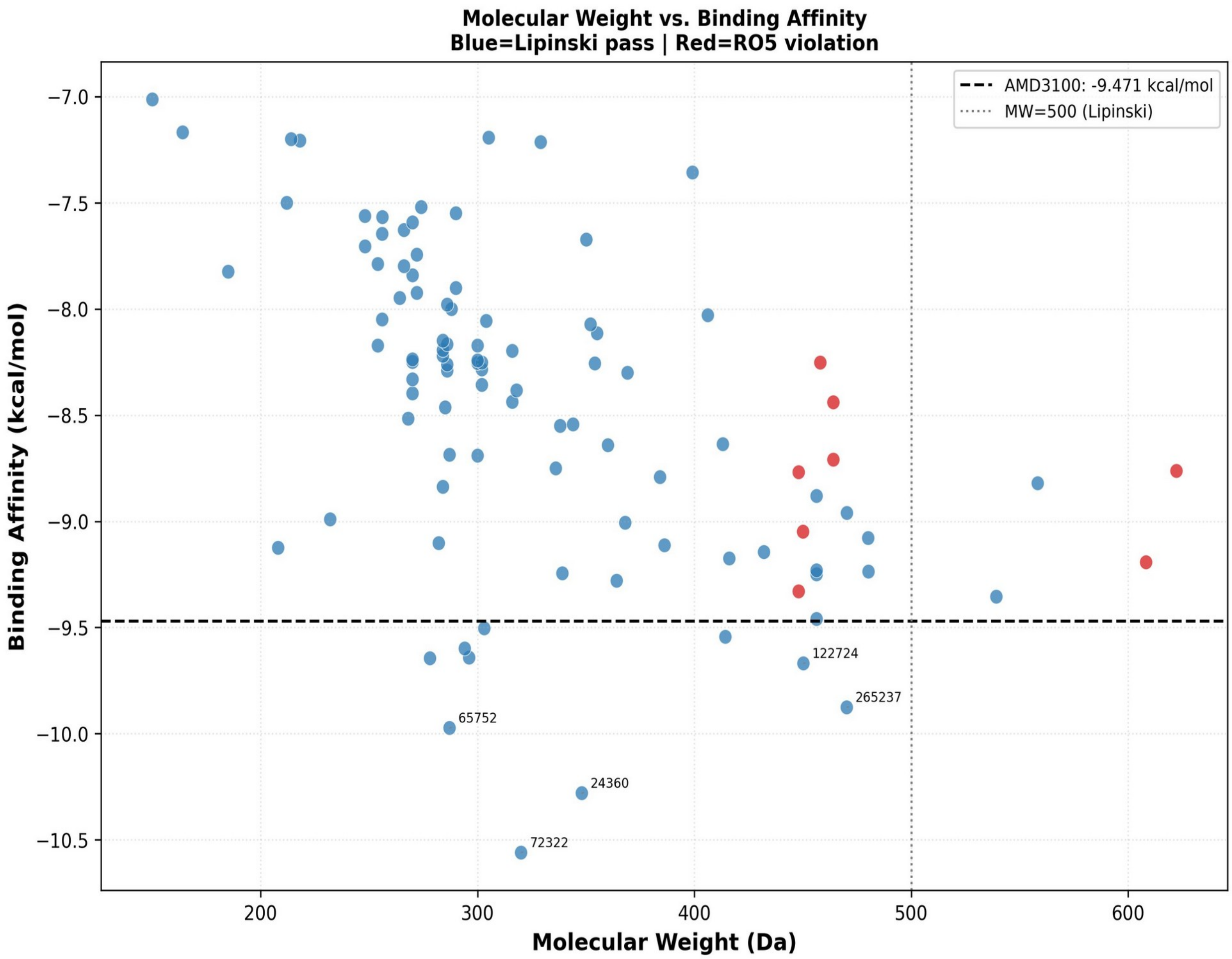

*(A) Distribution of AutoDock Vina scores. Red bars = compounds exceeding AMD3100 (−9.471 kcal/mol; dashed line). Left-skewed distribution confirms affinity enrichment in the library (10 hits, 9.9%).*

### B – Top 30 Compounds Ranked by Binding Affinity



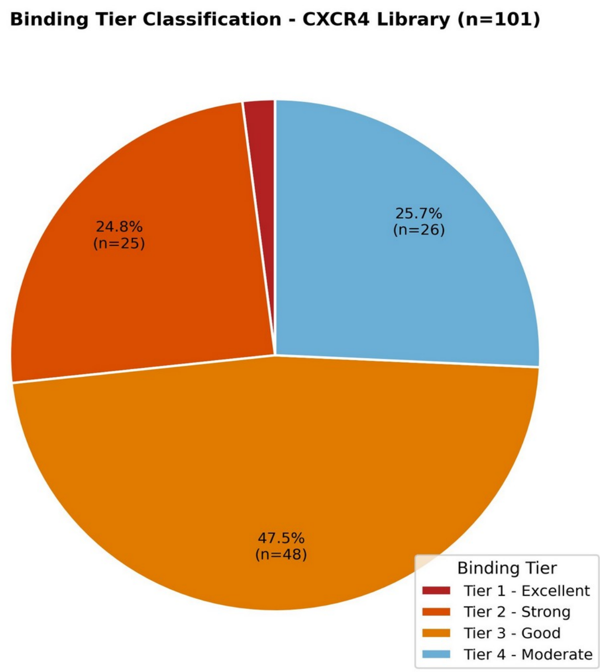

*(B) Ranked binding affinities of the top 30 compounds. Tier 1 (dark red) = exceeds AMD3100; Tier 2 (orange-red) = strong binders (−8.5 to −9.471 kcal/mol); Tier 3 (orange) = good binders (−8.0 to −8.5 kcal/mol).*

### C – Molecular Weight vs Binding Affinity



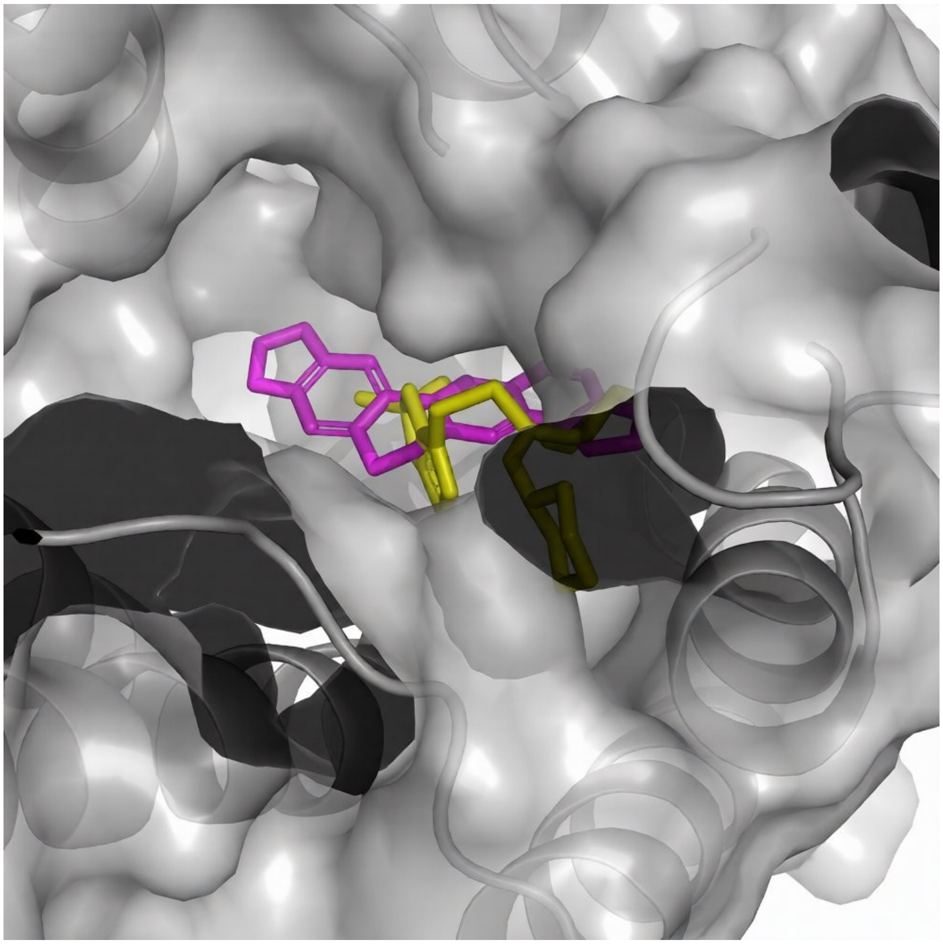

*(C) Molecular weight vs binding affinity. Blue = Lipinski pass; red = RO5 violation. Dashed line = AMD3100 threshold; dotted line = MW 500 Da (Lipinski limit). Top hits labelled.*

**Binding Tier Classification (n = 101)**



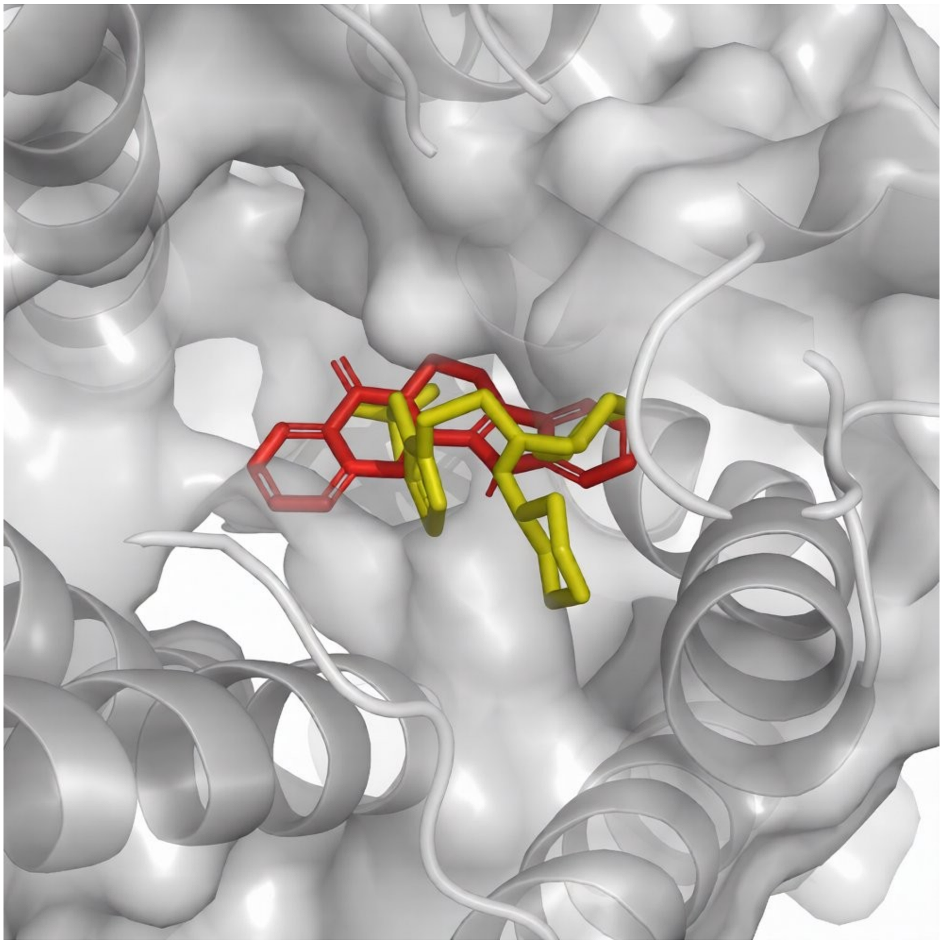

*(D) Binding tier classification: Tier 1 Excellent = 2.0% (n = 2); Tier 2 Strong = 24.8% (n = 25); Tier 3 Good = 47.5% (n = 48); Tier 4 Moderate = 25.7% (n = 26).*

**B – Coptisine (CID 72322; −10.560 kcal/mol)**



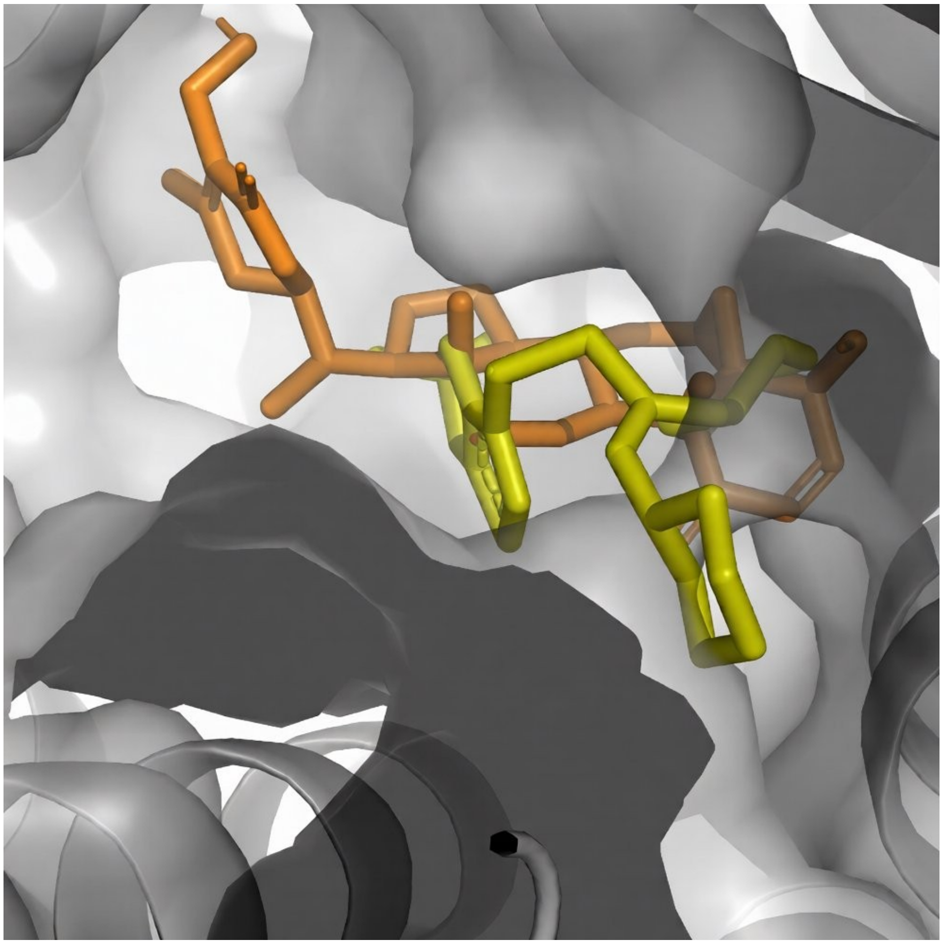
*(B) Coptisine (magenta) overlaid with IT1t co-crystal ligand (yellow) in the CXCR4 binding pocket. The isoquinoline core recapitulates the IT1t pose with π-stacking at Trp94 and Tyr116.*

**C – Genistein (CID 65752; −9.973 kcal/mol)**



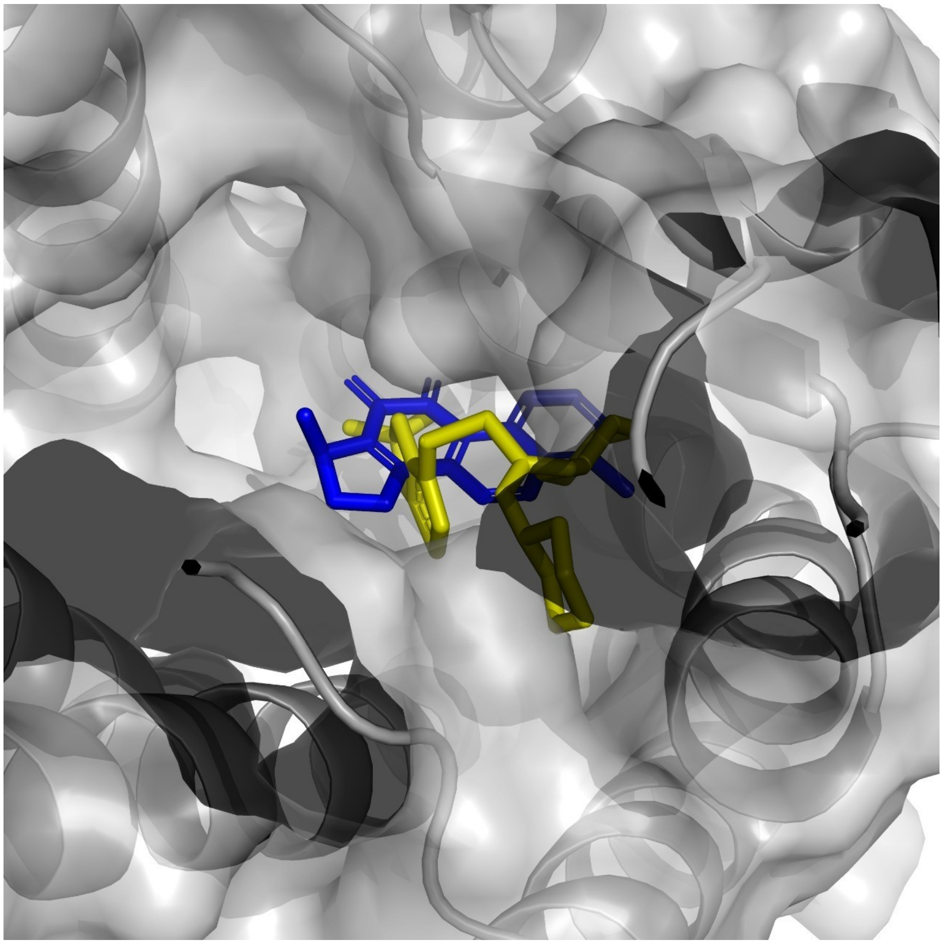
*(C) Genistein (red) in the CXCR4 binding pocket. A-ring hydroxyls engage Asp171/Glu288. Consistent with experimental CXCR4 inhibitory data reported by Park et al. (2014).[22]*

**D – Glycyrrhetinic Acid (CID 265237; −9.876 kcal/mol)**



*(D) Glycyrrhetinic acid (orange) in the TM4-TM5 hydrophobic channel. The 11-oxo group hydrogen-bonds with Tyr116. Anti-fibrotic activity in IPF-relevant models independently reported.[23]*

**E – Licochalcone A (CID 5316743; −9.645 kcal/mol)**

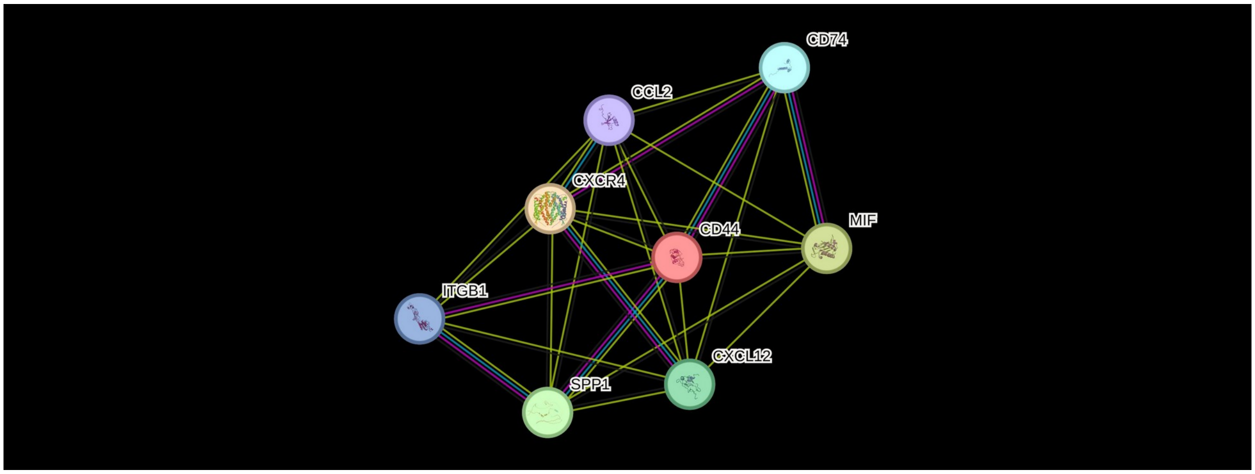


*(E) Licochalcone A (blue) with highest pose convergence (HIGH; spread = 1.676 kcal/mol). The α,β-unsaturated ketone is positioned near Cys109 in extracellular loop 1.[24]*

### 4.4 Network Pharmacology: CXCR4-Centred Hub Architecture

STRING PPI analysis[21] of eight hub proteins (CD44, CXCR4, MIF, SPP1, CXCL12, CD74, ITGB1, CCL2) produced a highly connected module (25 observed vs 5 expected edges; PPI enrichment $p = 2.27 \times 10^{-10}$), average clustering coefficient 0.912, average degree 6.25 (Figure 6). Functional enrichment revealed: cytokine activity (GO:0005125; FDR = 0.0069), integrin binding (GO:0005178; FDR = 0.044), ECM-receptor interaction (KEGG hsa04512; FDR = 0.0018), and leukocyte transendothelial migration (hsa04670; FDR = 0.0018). All eight proteins localise extracellularly, confirming pharmacological accessibility. The MIF-CD74-CXCR4 co-receptor complex (GO:0035692; FDR = 0.00059) mechanistically links the CellChat MIF pathway to the CXCR4 docking target, unifying all three analytical layers.

**Figure 4. STRING Protein-Protein Interaction Network of CXCR4 Hub Proteins**




*(Figure 6) STRING v12.0 PPI network of eight hub proteins convergent from CellChat and virtual screening analyses: CD44, CXCR4, MIF, SPP1, CXCL12, CD74, ITGB1, and CCL2. Network shows 25 observed edges vs 5 expected (PPI enrichment $p = 2.27 \times 10^{-10}$; confidence threshold ≥ 0.700). Line colours denote interaction evidence type: yellow = text-mining, cyan = database-curated, magenta = experimentally determined. All eight proteins localise extracellularly, confirming pharmacological accessibility without intracellular delivery requirements.*

## Discussion

This study presents an integrated three-layer computational framework combining scRNA-seq re-analysis, CellChat v2 cell-cell communication inference, and structure-based virtual screening of natural compounds against CXCR4, with network pharmacology validation. The convergence of all three analytical layers on the CXCL/CXCR4 axis as a central signalling hub in IPF constitutes the principal hypothesis-generating contribution of this work. All findings are in silico predictions requiring independent experimental validation.

### 5.1 Single-Cell Transcriptomics Confirms Known IPF Signatures

Our scRNA-seq analysis replicated key findings from the literature. The exclusive emergence of CTHRC1⁺ myofibroblasts in IPF replicates the landmark study by Adams et al. (2020).[4] The bimodal COL1A2 distribution (Figure 2B) indicates co-existence of quiescent and activated subpopulations – only the CTHRC1⁺ activated subpopulation likely drives active fibrosis. Dramatic epithelial loss (−67%) and fibroblast expansion (+65%) are consistent with the established epithelial-mesenchymal crosstalk model.[4,5,6] GO enrichment of IPF-upregulated fibroblast genes reflects the three pillars of IPF pathogenesis: matrix remodelling, myofibroblast apoptosis resistance, and ER stress.

### 5.2 CellChat Identifies Therapeutically Relevant Communication Rewiring

SPP1 was the most prominently gained pathway (score 10.0), consistent with published scRNA-seq studies identifying SPP1⁺ macrophages as primary drivers of fibroblast activation.[26,27] The CXCL pathway ranked third (score 9.0), with CXCL2/3/8 → ACKR1 showing the highest probability among newly gained IPF interactions, providing cellular and molecular evidence for CXCR4-axis activation in human IPF tissue.[8,9,10] The fibroblast autocrine self-loop (collagen-integrin and fibronectin-CD44 signalling) emerged as the dominant differentially gained interaction, identifying myofibroblast autocrine autonomy as a critical feature of advanced IPF.

### 5.3 Virtual Screening Prioritises Natural CXCR4 Binders

Of the four prioritised compounds, Coptisine is the most novel predicted CXCR4 binder – no prior CXCR4 activity has been reported for this isoquinoline alkaloid despite its known anti-inflammatory activity in lung injury models. Its highest docking score (−10.560 kcal/mol) and recapitulation of the IT1t crystallographic pose make it the primary candidate for radioligand binding validation. Genistein carries the strongest external support: published experimental CXCR4 inhibitory activity in cancer cell migration assays[22] provides independent evidence that the predicted binding mode is biologically plausible and not an artefact of the rigid-receptor model. Glycyrrhetinic acid and Licochalcone A both have documented anti-fibrotic activity in IPF-relevant models,[23,24] though their reported mechanisms (TGF-β/Smad2/3 inhibition) may operate in parallel to CXCR4 antagonism, potentially conferring dual therapeutic benefit. Licochalcone A's HIGH pose convergence (spread = 1.676 kcal/mol) is particularly notable: tightly clustered binding modes indicate a well-defined binding geometry less likely to represent a docking artefact, strengthening confidence in the predicted interaction.

### 5.4 Limitations

No experimental validation – all docking predictions, CellChat inferences, and network pharmacology findings are in silico and require confirmation via CXCR4 radioligand displacement assays, IPF fibroblast functional assays, and in vivo models. Small scRNA-seq cohort (n = 3 per group) – proportional differences may not generalise; formal compositional testing (e.g., scCODA) requires a larger cohort. Rigid receptor docking – AutoDock Vina does not capture CXCR4 conformational flexibility; molecular dynamics simulations would strengthen binding mode predictions. Limited ADMET profiling (Lipinski RO5 only). Potential promiscuous binders or aggregate-based false positives that can only be ruled out experimentally.

### 5.5 Proposed Experimental Validation Workflow

Tier 1 (essential): Radioligand displacement ($IC_{50}$ for top 3 compounds); IPF fibroblast proliferation assay (with/without TGF-β stimulation). Tier 2 (supporting): β-arrestin recruitment assay; CXCL12-induced fibroblast migration; collagen gel contraction assay. Tier 3 (in vivo): bleomycin-induced pulmonary fibrosis in mice (hydroxyproline content, Ashcroft score, qPCR for Col1a1/Acta2, bronchoalveolar lavage cell differentials).

## Conclusion

This in silico study demonstrates three main findings: (1) The CXCL/CXCR4 signalling axis is among the most prominently activated intercellular communication pathways in human IPF lung tissue, as inferred from CellChat v2 analysis of scRNA-seq data (GSE132771).[16,17] (2) Four natural compounds – Coptisine, Genistein, Glycyrrhetinic acid, and Licochalcone A – are computationally predicted to bind CXCR4 with affinity exceeding AMD3100, based on AutoDock Vina virtual screening against PDB: 3ODU.[18,19,20] (3) Network pharmacology confirms that CXCR4 and its hub partners form a highly interconnected, extracellularly accessible signalling complex.[21] All predictions require experimental validation via CXCR4 radioligand displacement assays, primary IPF fibroblast functional assays, and preclinical in vivo models.

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